Leak or Keep?
Learning to Share Interview Questions Among Competing Candidates

Final Project Report — Multi-Agent Systems, IIT Gandhinagar

Jash Bharat Pasad 24110146 jash.pasad@iitgn.ac.in
Bhavay Goyal 24110070 bhavay.goyal@iitgn.ac.in
Chaitanya Bhoite 24110086 chaitanya.bhoite@iitgn.ac.in
Vishnu Sinha 24110389 vishnu.sinha@iitgn.ac.in

Code, result files and commands that regenerate every number below:
https://github.com/Vish-Overflows/multiagent-systems-leak-or-keep

Abstract. After an interview, a candidate can tell friends who interview later which questions were asked. This helps them, but they may be competing for the same offers. We model this as a game on an $n \times n$ sharing matrix: candidates interview in a fixed order, know only a coarse ability type (never their score), and a shared question may not reappear. We solve the six-candidate game *exactly* for 378 combinations of hiring rule, friendliness, friendship network and leak model, and then let bandit learners who know none of the rules play it for 200,000 rounds (900 runs). Four findings. (1) *The hiring rule decides who talks.* If the company hires everyone above a cut-off, everybody tells as soon as they care about friends at all. If it fills a fixed number of seats, only candidates with little to lose tell: weak ones first (once they value a friend's offer at 0.10 of their own), strong ones last (0.38), and a candidate who is friends with everyone never. (2) Disliking outsiders makes candidates tell *more* under fixed seats and changes nothing under a cut-off, as our theory predicts. (3) Learners who see only their own outcome find this equilibrium: UCB1 ends with 93–99% of decisions correct, and a learner that uses one fact about leaks (a tell that did not land is the same as no tell) gets there two to three times sooner. (4) Leaks cost a cut-off company 8–12 times more than a fixed-seats company: it needs a bank of 40 questions, against 9, to keep the loss under 5%.

Keywords Bayesian games · multi-agent bandits · empirical game theory · coarse correlated equilibrium

1 Introduction

Every placement season, a candidate leaves the interview room and within minutes their friends know what was asked. It feels like helping friends, but when offers are limited, a friend helped may take the helper's seat. Leaks happen at scale [1, 2]; industry treats them as something that happens to questions (bigger banks, rotation [2]), not as a choice made by competing candidates.

Example. Two offers; Aarav, Meera, Kabir and Diya interview in that order, and Kabir and Diya are Aarav's friends. Each answers 3 of 10 bank questions, and knowing a question adds 4 of its 10 marks. With scores 21, 22, 20 and 19, Meera and Aarav are hired. If Aarav shares, each friend sees at least one of his questions with probability $1 - \binom{7}{3}/\binom{10}{3} \approx 0.71$, and with probability 0.92 one of them overtakes him. If the company instead hires everyone scoring at least 20, sharing costs Aarav nothing — and Diya, meant to be rejected, gets in with probability 0.71.

Question. Who shares with whom, how does this depend on the hiring rule and on what candidates care about, and can candidates who do not know the payoffs *learn* to share as the equilibrium predicts?

Related work. Rank-order versus absolute rewards [3] and helping or sabotaging rivals [4] are well studied, but there help is effort, not information. Li and Conitzer [5] model leaked test questions as a Stackelberg game with the leak given; we make it a choice among competitors. With unknown payoffs, candidates are bandit learners [6, 7, 8], whose no-regret play approaches coarse correlated equilibria [9]; solving a simulated game is empirical game theory [10].

What this report adds to the project review. The review fixed the model, proved Lemma 1 and Propositions 1–3, and showed a one-setting pilot. Here we describe the algorithms (Section 4), report all four planned experiments (Section 5), and give a verdict on each of the five hypotheses fixed in advance (Table 4). We use two plain names: *fixed seats* (Seats: the best K scores are hired) and *cut-off* (Bar: everyone above a fixed score is hired).

2 Preliminaries

Notation. $N = \{1, \dots, n\}$ are the candidates, numbered in their (fixed) interview order. A symmetric matrix $W \in \{0,1\}^{n \times n}$ records friendships: $F(i)$ is the set of i's friends and $L(i) = \{j \in F(i) : j > i\}$ those who interview after i. Candidate i has a skill s_i and a coarse type $\theta_i \in \Theta = \{\text{weak}, \text{average}, \text{strong}\}$, the tercile of s_i. The company has K seats, a bank of Q questions, and asks m per interview. Y_j is j's score, H the hired set, and $x_j = \mathbf{1}[j \in H]$. The sharing matrix $S \in \{0,1\}^{n \times n}$ has $S_{ij} = 1$ if i tells j; row $S_{i\cdot}$ is i's action.

Definition 1 (Bayesian game and strategy)**.** A Bayesian game is a tuple $\langle N, (A_i), (\Theta_i), p, (u_i) \rangle$: players, action sets, type sets, a common prior p over types, and utilities u_i depending on all actions and types [11]. A strategy is a map $\sigma_i : \Theta_i \to \Delta(A_i)$; it is pure if it never randomises. *A strategy is a rule "given my type, tell these friends".*

Definition 2 (Bayes–Nash equilibrium, BNE)**.** A profile σ is a BNE if for every player i, type θ_i and action a_i', $\mathbb{E}[u_i(\sigma_i(\theta_i), \sigma_{-i}) \mid \theta_i] \geq \mathbb{E}[u_i(a_i', \sigma_{-i}) \mid \theta_i]$. The *exploitability* of σ is the largest expected gain any type can obtain

by deviating; σ is an ε-BNE if its exploitability is at most ε. *Nobody, whatever their type, can do better by changing only their own rule.*

Definition 3 (Coarse correlated equilibrium, CCE)**.** A distribution z over strategy profiles is a CCE if for every player i and every fixed alternative σ_i', $\mathbb{E}_{\sigma\sim z}[U_i(\sigma)] \geq \mathbb{E}_{\sigma\sim z}[U_i(\sigma_i', \sigma_{-i})]$, where U_i is expected utility over types and the environment. Every BNE is a CCE, and the CCEs form a polytope, so linear programming finds them. *CCEs are what players who "have no regrets" can end up playing, so they are the right yardstick for learners.*

Definition 4 (Contextual bandit and regret)**.** In round t a learner sees a context x_t (here her type), picks an arm a_t (a row of S), and observes only the reward r_t of that arm. Its regret after T rounds is $\max_\pi \sum_{t\leq T} r_t(\pi(x_t)) - \sum_{t\leq T} r_t(a_t)$, the reward lost against the best fixed rule π from contexts to arms. *The learner never sees what the other choices would have paid.*

3 Problem Formulation

Stage game $\Gamma_{R,\phi,W,E}$, indexed by the hiring rule R, the utility type ϕ, the network W and the leak model E.

- *Order and types.* $s_i \overset{iid}{\sim} \mathcal{N}(0,1)$; θ_i is the tercile of s_i, and i observes only θ_i.
- *Action.* $S_{i\cdot} \in \{0,1\}^n$ with $S_{ij} = 0$ unless $j \in L(i)$: a candidate can tell only friends who interview after her. With two friend groups $\{1,3,5\}$ and $\{2,4,6\}$ exactly six entries are free: $S_{13}, S_{15}, S_{35}, S_{24}, S_{26}, S_{46}$.
- *Leaks.* Let $n_j = \sum_{i<j} S_{ij}$ be the number of people who tell j, and k_j the number of j's m questions seen in advance. **E0**: $k_j = m\,\mathbf{1}[n_j > 0]$ (a told question always comes up). **E1**: $k_j \sim \mathrm{Bin}\big(m, 1-(1-m/Q)^{n_j}\big)$ (comes up by chance). **E2**: $k_j = |A_j \cap \bigcup_{i:S_{ij}=1} A_i|$ with $A_j \sim \mathrm{Unif}\binom{[Q]}{m}$ (actual questions drawn from the bank).
- *Score.* $Y_j = s_j + \frac{1}{m}\big(\Delta k_j + \sum_{q=1}^{m} \eta_{jq}\big)$, $\eta_{jq} \sim \mathcal{N}(0,\sigma^2)$: skill, plus a bonus Δ per question seen in advance, plus marking noise. Candidates never observe Y_j.
- *Hiring.* SEATS: $H = \text{top-}K(Y)$. BAR: $H = \{j : Y_j \geq \tau_K\}$ with $\tau_K = \sqrt{1+\sigma^2/m}\;\Phi^{-1}(1-K/n)$, the cut-off that hires K people on average when nothing leaks.
- *Utilities*, with $|S_{i\cdot}|$ the number of friends told and c a small cost per tell: $u_i^{\text{SELF}} = x_i - c|S_{i\cdot}|$; $u_i^{\text{FRIEND}(\alpha)} = u_i^{\text{SELF}} + \alpha\sum_{j\in F(i)} x_j$; $u_i^{\text{CLAN}(\alpha,\beta)} = u_i^{\text{FRIEND}(\alpha)} - \beta\sum_{j\notin F(i)\cup\{i\}} x_j$; $u_i^{\text{FAIR}(a,b)} = u_i^{\text{SELF}} - \frac{1}{n-1}\sum_j \big[a(x_j-x_i)^+ + b(x_i-x_j)^+\big]$ [12]. *Selfish candidates want only their own offer; friendly ones also value a friend's offer at α of their own; clannish ones dislike outsiders' offers; fairness-minded ones dislike being behind (a) or ahead (b).*

Level 1 (known rules): find the equilibria of Γ. **Level 2 (unknown rules):** rounds $t = 1,\dots,T$ are i.i.d. copies of Γ; candidate i knows her own utility function and her arms, but not E, (Q,m,Δ,σ) or anyone else's utility or strategy; she observes θ_i^t, chooses $S_{i\cdot}^t$ and sees only u_i^t.

Research questions. RQ1: the equilibrium $S^\star$ and the break-even friendliness $\alpha^\star_{i,\theta}$ (the smallest α at which type θ of candidate i tells), by position, type, W and R. RQ2: how utility types and mixed populations change it. RQ3: for each learner, regret, agreement with the equilibrium and CCE gap. RQ4: the company's hiring-value loss $\Lambda_R = \mathbb{E}[V \mid \text{no leaks}] - \mathbb{E}[V]$, $V = \sum_{j\in H}(s_j - \tau_K)$, as a function of Q.

Structural results. Let ℓ_i be i's own loss in hiring probability from sharing, and G_i the gain in her friends' expected hires.

Lemma 1. *With all random draws and all other rows fixed, setting $S_{ij} = 1$ weakly raises Y_j and changes no other score. Hence $\ell_i \geq 0$ and $G_i \geq 0$ under both rules.*

Proposition 1 (No sharing in a clique)**.** *With SEATS, complete W, FRIEND(α), $\alpha \leq 1$ and $c > 0$, no one shares in any BNE.*

Proof. $|H| = K$, so $|F(i)\cap H| = K - x_i$ and $u_i = \alpha K + (1-\alpha)x_i - c|S_{i\cdot}|$. Sharing weakly lowers x_i (Lemma 1) and costs c. □

Corollary 1 (The popular candidate stays silent)**.** *The proof uses only $F(i) = N\setminus\{i\}$. So under SEATS a candidate who is friends with* everyone *never shares, whatever the rest of the network looks like and whatever the others do.*

Proposition 2 (Sharing is free under a cut-off)**.** *Under BAR and $\phi \in \{\text{SELF}, \text{FRIEND}, \text{CLAN}\}$, $\ell_i \equiv 0$ and the gain from any row does not depend on θ_i, so each candidate has a best response that is the same for all her types. (For FAIR this can fail: the penalty depends on whether i herself is hired.)*

Proposition 3 (Leak formula)**.** *In E2, each of j's questions is leaked with probability exactly $1-(1-m/Q)^{n_j}$. The expected leaked fraction is at most ε if $Q \geq m/(1-(1-\varepsilon)^{1/n_j})$, and $Q \geq mn_j/\varepsilon$ always suffices.*

Proofs. Lemma 1: Y_j is monotone in k_j, and k_j in n_j. Prop. 2: $x_i = \mathbf{1}[Y_i \geq \tau_K]$ does not depend on S. Prop. 3: independent uniform sets and $(1-\varepsilon)^{1/n} \leq 1 - \varepsilon/n$. *Remark (clans).* Under SEATS, $|H| = K$ gives $u_i^{\text{CLAN}(\alpha,\beta)} =$

$(1+\beta)x_i + (\alpha+\beta)\sum_{F(i)} x_j - \beta K - c|S_i.|$: up to a constant, a friendly candidate with $\alpha_{\text{eff}} = (\alpha+\beta)/(1+\beta) > \alpha$. Disliking outsiders should therefore *raise* sharing under SEATS and do nothing under BAR.

4 Methodology

The pipeline has three stages: simulate the game into a table, solve the table (Level 1), and let learners play the simulator (Level 2). All experiments use $n = 6$, $K = 2$, $m = 3$, $Q = 15$, $\Delta = \sigma = 1$, $c = 0.001$ unless a parameter is being swept.

4.1 Stage 1: from simulator to payoff table

For each of the $3^6 = 729$ type profiles θ and each of the 2^f sharing matrices S (f free entries; $f = 6$, i.e. 64 matrices, for two friend groups) we simulate 2,000 interview days and store $P(x_j = 1 \mid \theta, S)$, $P(x_i x_j = 1 \mid \theta, S)$ and $\mathbb{E}[V \mid \theta, S]$ for both hiring rules. Three choices make this cheap and precise:

- *Same days for every S.* Skills, marking noise and leak draws are generated once per day and reused for all matrices and both rules (common random numbers). Payoff *differences* between two sharing choices — all that equilibria depend on — are then far more precise than the payoffs themselves, and Lemma 1 holds exactly in the table, not just on average.
- *Stratify by type.* Skills are drawn from the normal distribution truncated to each candidate's tercile, so all 729 profiles get the same sample size.
- *Simulate once, re-use for every utility.* All four utility families are linear in the stored probabilities (FAIR needs the pairwise ones, since $(x_j - x_i)^+ = x_j - x_i x_j$). Changing α, β, (a, b) or mixing utility types needs no new simulation.

One table for two friend groups (93 million simulated outcomes) takes 15–40 s on a laptop. We also store the means over 20 blocks of 100 days, used for the bootstrap below.

4.2 Stage 2 (Level 1): solving the known game

A pure strategy of candidate i assigns one of her $2^{|L(i)|}$ rows to each of her three types. With two friend groups this gives $4^3 \cdot 4^3 \cdot 2^3 \cdot 2^3 = 262{,}144$ profiles. The central object is the *interim value*

$$V_i[\sigma_{-i}, \theta_i, a] = \mathbb{E}_{\theta_{-i}}\big[u_i\big(\theta,\, S(a, \sigma_{-i}(\theta_{-i}))\big)\big],$$

the expected utility of type θ_i of candidate i from row a when the others follow σ_{-i}. We precompute it for every σ_{-i}; everything else is a lookup.

- **Pure equilibria: exhaustive.** For every profile we compute the exploitability $\max_{i,\theta_i}\big(\max_a V_i[\sigma_{-i}, \theta_i, a] - V_i[\sigma_{-i}, \theta_i, \sigma_i(\theta_i)]\big)$ and keep the profiles with value 0. The list of pure BNEs is therefore complete, not a sample from a search.
- **Independent checks.** Best-response dynamics from 8 starting profiles (nobody shares, everybody shares, 6 random) must end in an enumerated equilibrium. To look for mixed equilibria we run replicator dynamics on the 12 (candidate, type) agents from 3 random interior starts.
- **All CCEs at once.** A CCE is a distribution over the profiles satisfying one linear inequality per (candidate, alternative strategy) pair. We solve two linear programs (HiGHS): maximise and minimise the expected share of tells. If the two optima coincide, *every* CCE — hence every mixed equilibrium and every limit of no-regret learning — produces the same amount of sharing.
- **How sure are we? Bootstrap ε.** The table is an estimate. We resample its 20 blocks of days with replacement (200 times; 100 for the two smaller networks), rebuild the utilities, and recompute the exploitability of the reported equilibrium. We report the share of replicates in which it is still an exact equilibrium, and the 95th percentile of ε.
- **Break-even friendliness** is measured twice: by the formula $\alpha^\star = (\ell_i + c)/G_i$ from the table, and directly as the smallest α on a grid of step 0.025 at which that type tells in every equilibrium.

The complete network has 15 free entries (2^{15} matrices; too many profiles to enumerate at $n = 6$). There we solve $n = 4$ exhaustively and, at $n = 6$, test the two candidate profiles "nobody shares" and "everybody shares" against every deviation of every type of every candidate.

4.3 Stage 3 (Level 2): learning the unknown game

Each (candidate, type) pair is one bandit learner whose arms are that candidate's rows; with two friend groups there are 12 learners with 4 or 2 arms. Each round draws fresh skills, the realised types' learners choose, the simulator runs one day, and each learner receives only its own realised utility (rescaled to $[0, 1]$). We compare five learners:

- **ε-greedy**: explore uniformly with probability $\varepsilon_t = t^{-1/3}$, else play the best arm so far. **UCB1** [6]: play $\arg\max_a \hat{\mu}_a + \sqrt{2 \ln t / n_a}$. **Thompson sampling** [8]: sample each arm's mean from a Gaussian posterior, play

Algorithm 1: One round of Leak-aware UCB for candidate i of type θ (arms $a \subseteq L(i)$)

play $a = \arg\max_{a'} \ \hat{\mu}_{\theta,a'} + \sqrt{2 \ln t_\theta / n_{\theta,a'}}$; observe own utility r and, for each told friend $j \in a$, the number $k_j^{(i)}$ of told questions that came up;
$D \leftarrow \{j \in a : k_j^{(i)} = 0\}$; `// tells that did not land`
for *every* a' *with* $a \setminus D \subseteq a' \subseteq a$ **do**
  $n_{\theta,a'} \leftarrow n_{\theta,a'} + 1$; $\hat{\mu}_{\theta,a'} \leftarrow$ running mean with the new sample $r + c\,|a \setminus a'|$; `// refund cost of unmade tells`
update leak-rate estimate $\hat{p} \leftarrow$ mean of $k_j^{(i)}/m$ over all tells so far; $t_\theta \leftarrow t_\theta + 1$;

Table 1: Verification checks. "Deviations" are all rows of every type of every candidate, against both "nobody shares" and "everybody shares".

Check	What must hold	Simulator	Verdict
Worked example	0.7083, 0.9149, 0.7083 (exact)	0.7085, 0.9157, 0.7088	pass
Lemma 1	a tell never lowers the friend's score, changes no one else's	min change 0, others 0	pass
Prop. 3	leaked fraction $= 1-(1-m/Q)^{n_j}$, $n_j = 1,2,3$	largest error 0.0013 (2.2 s.e.)	pass
Prop. 2	under a cut-off, own row never changes own hire	largest change 0	pass
	... and friends' gain is the same for every own type	largest spread 0.0029	pass
Prop. 1	everyone friends, fixed seats: nobody tells (n=4 solved, n=6 deviations)	unique equilibrium; 0 of 684 deviations pay	pass
Selfish baseline	nobody tells at $\alpha = 0$ (both rules, E0–E2)	6 of 6 games	pass

the largest. **EXP3** [7]: exponential weights on importance-weighted rewards; assumes nothing about how rewards are generated.

- **Leak-aware UCB** (ours; Algorithm 1). The review promised a "model-based" learner that estimates the leak rate and Δ and best-responds in its estimated game. Best-responding needs the other candidates' strategies, which nobody observes, so we kept the idea — *learn how the environment works, not only what each arm pays* — in a form that does not. It uses one fact: *a tell matters only through questions that actually come up.* If my friend reports that none of the questions I told her appeared, the day went exactly as if I had not told her, so the same observation is also a valid sample for the arm "did not tell her" (plus the small cost c I would have saved). In E1/E2 about half of all tells do not land ($0.8^3 = 0.51$), so most pulls of a "tell" arm also teach about the "silent" arms. The price is one extra piece of feedback: how many of my questions came up for my friend. Under E0 every tell lands, and the learner is identical to UCB1.

What we measure. *Agreement*: the share of the 12 learners whose currently-best arm is the equilibrium row (if a game has several equilibria, the best-matching one). *Regret*: Definition 4, computed with the table's expected payoffs so that luck does not enter. *CCE gap*: the largest average gain any candidate could have made by switching to one fixed rule for the whole run (distance of the history of play from a CCE). *Rounds to 90%*: first round at which agreement reaches 0.9. *Residual gain*: the exploitability of the profile of most-played arms — what anyone could still gain by deviating from where the learners settled.

Experiments and statistics. X1: rule $\times$ $\alpha \in \{0, 0.025, \ldots, 0.5\}$ $\times$ four networks $\times$ E0–E2. X2: four utility families and five mixed populations. X3: five learners $\times$ two rules $\times$ E0–E2, $T = 200{,}000$ rounds, 30 seeds each. X4: bank size $Q \in \{4, \ldots, 64\}$ in E2, equilibrium re-solved for each Q. Runs that share a seed share every random draw of the environment across rules, leak models and learners, so comparisons are paired. We report 95% bootstrap confidence intervals over seeds, two-sided sign-flip permutation tests on the paired differences, and Holm-corrected p-values within each family of tests [13].

Checks before any experiment (Table 1). The simulator must reproduce the worked example and every structural result of Section 3; all pass. A further 27 unit tests cover every module.

5 Results

5.1 Who tells? The hiring rule decides (RQ1)

Under fixed seats only weak candidates tell; under a cut-off everyone does (Figure 1). The pilot's picture survives exhaustive search: with fixed seats 33% of the possible tells are made, with a cut-off 100%. Under both rules the equilibrium is unique, all 8 best-response runs and all 3 replicator runs end in it, and it remains an exact equilibrium in 100% (seats) and 100% (cut-off) of 200 bootstrap replicates of the table.

Why: a weak candidate has almost nothing to lose (Figure 2a,b). Under fixed seats, telling one friend costs a weak candidate 0.38 percentage points of her own hiring chance, an average one 1.49 and a strong one 1.60 — a weak candidate is rarely near the last seat, so a boosted friend rarely pushes *her* out. The friend gains about the same in all three cases (4.5, 4.9, 4.5 points). The break-even friendliness $\alpha^\star = (\ell + c)/G$ is therefore 0.10, 0.32 and 0.38 for weak, average and strong candidates (the pilot estimated 0.09, 0.32, 0.37), and the exact equilibria switch at the next grid points: 0.125, 0.325–0.35 and 0.375–0.4. Under a cut-off the loss is exactly zero for every type (Proposition 2)

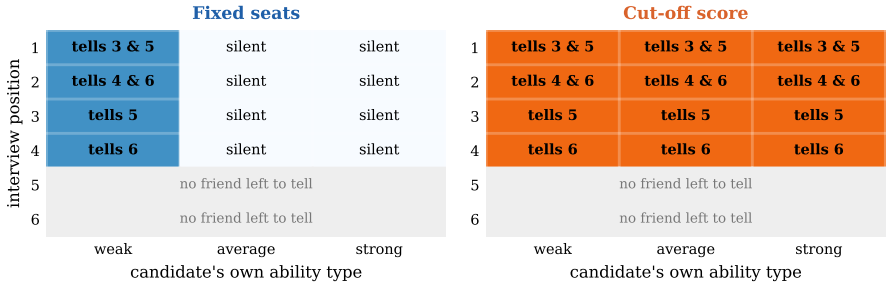


Figure 1: The equilibrium for two friend groups $\{1, 3, 5\}$, $\{2, 4, 6\}$ at friendliness $\alpha = 0.2$ (leak model E1). Under either rule it is the only pure equilibrium among 262,144 profiles.

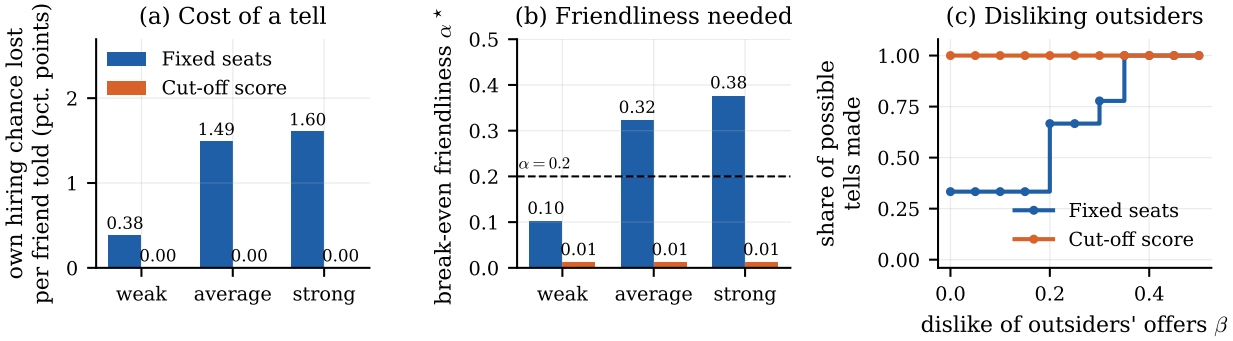


Figure 2: (a) Own hiring chance lost per friend told. (b) Break-even friendliness; types below the dashed line tell at $\alpha = 0.2$. (c) Share of possible tells made in equilibrium as candidates dislike outsiders' offers more ($\alpha = 0.2$). Two friend groups, E1.

and $\alpha^\star \approx 0.01$: any friendliness above the cost of a message is enough.

The staircase holds across networks and leak models (Figure 3). We solved 378 games exactly. Under a cut-off, sharing jumps from nothing to its maximum at the first grid point above zero in every network. Under fixed seats it climbs in steps — weak, then average, then strong. A cut-off produces at least as much sharing as fixed seats in 188 of 189 comparisons, and strictly more in 134; the one exception is discussed below. Three details:

- *The popular candidate never tells.* In the star network the hub is friends with everyone. Her break-even friendliness is above 1 and she is silent at every $\alpha \leq 0.5$ under fixed seats, as Corollary 1 says: any seat she hands a friend is taken from herself or from another friend. Sharing in the star therefore stops at 40% (the two candidates before her).
- *Everyone friends: nobody tells* under fixed seats ("nobody shares" passes the deviation test at all 21 values of α, "everybody shares" at 0), and everybody tells under a cut-off (at 19 of the 20 values $\alpha > 0$; at the smallest, a few tells are not worth the message cost). This is Proposition 1 at $n = 6$.
- *When told questions always come up (E0), a second tell is wasted.* A candidate needs to be told only once, so equilibria differ in *who* tells her, and sharing tops out below 100% (4 of 6 tells in two groups). This is the only place we find several equilibria: 63 of 378 games, all in E0; the other 315 have exactly one and none has zero. It is also where the one exception sits (two friend groups, E0, $\alpha = 0.375$): fixed seats have 8 equilibria sharing 61–78%, the cut-off 4 sharing 67%.

How solid are these equilibria? The reported profile stays an exact equilibrium in at least 95% of bootstrap replicates in 370 of 378 games (98%). The exceptions sit on a step of the staircase, where a type is nearly indifferent; even there the 95th-percentile ε is at most 0.03 percentage points of utility. So the *pattern* is certain and each *threshold* is known to within one grid step. Best-response dynamics converged in 3024 of 3024 runs, and replicator dynamics ended at an enumerated pure equilibrium in 375 of 378 runs (the other 3, all in the most fragile game, had not settled after 3,000 steps). The CCE linear programs agree from the other side: maximum and minimum sharing over *all* CCEs coincide in 106 of 106 games, so no mixed or correlated equilibrium shares more or less than the pure one.

5.2 What candidates care about (RQ2)

Disliking outsiders makes candidates more generous to friends — but only under fixed seats (Figure 2c). As β rises from 0 to 0.5 at fixed $\alpha = 0.2$, sharing under fixed seats rises from 33% to 100% (everyone tells from $\beta = 0.35$, i.e. $\alpha_{\text{eff}} = 0.41$, in line with the strong types' threshold above), and stays at 100% under a cut-off. This is the Remark of Section 3: with a fixed number of seats, each seat a friend takes is one an outsider loses.

Fairness-minded candidates behave the other way round (Table 2). Under fixed seats they never tell. Under

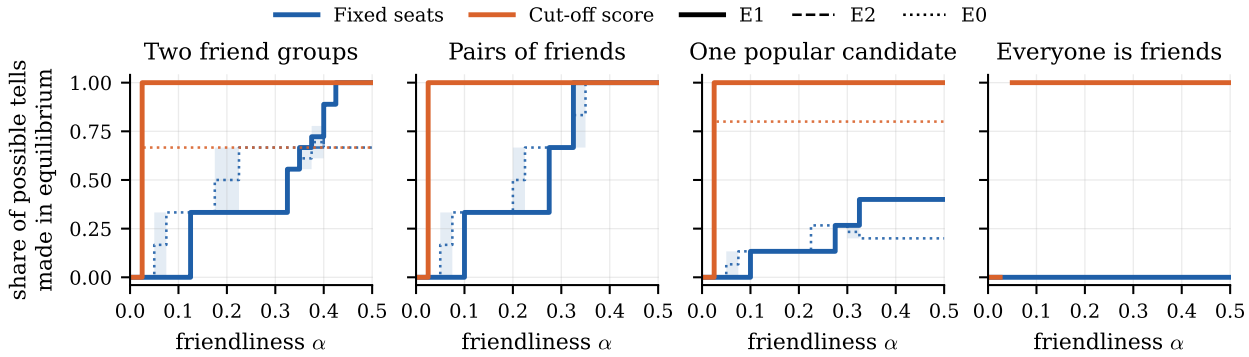


Figure 3: Share of possible tells made in equilibrium as candidates care more about friends. Shaded bands: range over multiple equilibria (E0 only). "Everyone is friends" shows the profile that passes the deviation test (E1).

Table 2: Equilibrium sharing for different motives (two friend groups A= $\{1,3,5\}$, B= $\{2,4,6\}$; E1). Every game has exactly one pure equilibrium. †: exact equilibrium in fewer than 95% of bootstrap replicates (a type is nearly indifferent).

What candidates care about	Who tells, in equilibrium		Tells made (%)	
	Fixed seats	Cut-off	Seats	Cut-off
Only own offer (selfish)	nobody	nobody	0	0
Friends' offers too, α=0.2	weak	everybody	33	100
... and dislike outsiders, β=0.2	weak; average†	everybody	67	100
... and dislike outsiders, β=0.4	everybody	everybody	100	100
Dislike inequality, (a,b)=$(0.25, 0.1)$	nobody	nobody	0	0
Dislike inequality, (a,b)=$(0.5, 0.25)$	nobody	strong (4 of 6 tells)†	0	22
Dislike inequality, (a,b)=$(1, 0.5)$	nobody	strong	0	33
Dislike inequality, (a,b)=$(2, 0.5)$	nobody	nobody	0	0
Group A friendly, group B selfish	weak in A	everyone in A	17	50
Only the first interviewees friendly	weak (cand. 1–2)	everyone (cand. 1–2)	22	67
Only the first interviewees selfish	weak (cand. 3–4)	everyone (cand. 3–4)	11	33
Group A friendly, group B clannish (β=0.3)	weak; average in B; strong (2 of 6 tells)†	everybody	61	100
Group A inequality-averse, group B friendly	weak in B	weak in B; average in B; strong (5 of 6 tells)	17	61

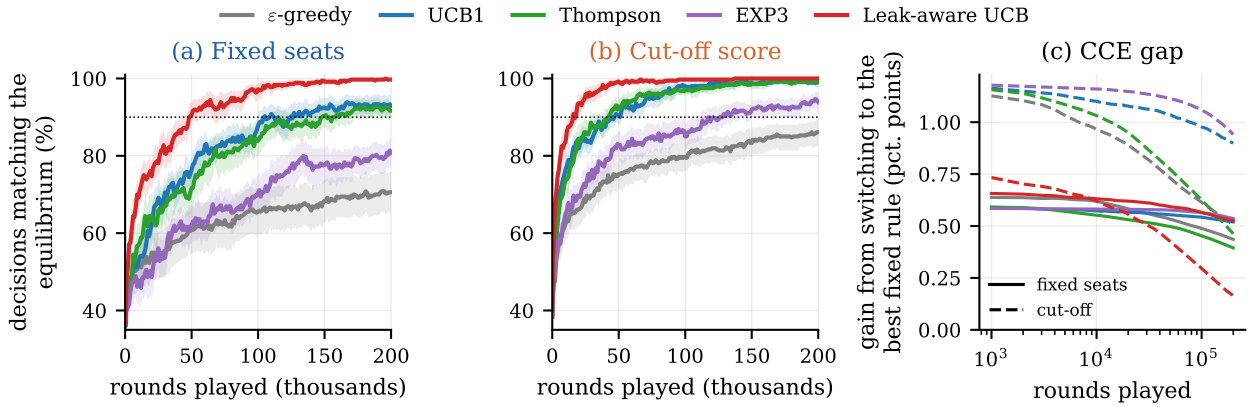


Figure 4: Learning in E1, mean over 30 seeds with 95% bootstrap bands. (a,b) Share of the 12 (candidate, type) learners whose best arm is the equilibrium row. (c) CCE gap of the history of play.

a cut-off it is the *strong* who tell, and only for moderate inequality aversion: a strong candidate expects to be hired and dislikes being ahead of her friends, so she pulls them up; if envy (a) is large, the risk of being the one left behind dominates and telling stops. This is the case Proposition 2 excludes, and the only one where behaviour under a cut-off depends on type.

Motives barely spill over. In the 10 mixed-population games, only 1 of 120 (candidate, type) choices differs from what that candidate does in a population made entirely of her own kind (a nearly indifferent strong clannish candidate tells one friend more when the other group is merely friendly). With one friendly and one selfish group, exactly the friendly group shares: weak types under fixed seats, everyone under a cut-off.

5.3 Can candidates learn it? (RQ3)

Yes — but the signal is tiny, so it takes tens of thousands of interviews. At equilibrium the best row beats the second-best by only 0.36–0.80 percentage points of utility under fixed seats and 0.96–0.99 under a cut-off, while a single outcome is 0 or 1. UCB1 and Thompson sampling nevertheless end with 93% and 92% (seats) and 99% and 99% (cut-off) of decisions correct in E1 (Table 3); EXP3, which guards against adversarial rewards, and ε-greedy are

Table 3: Decisions matching the equilibrium after 200,000 rounds (%, mean ± half-width of the 95% CI over 30 seeds; best per column in bold), and median rounds to reach 90% ("never (x%)": the median run did not, x% of runs did). In E0 with a cut-off the equilibria differ only in who makes a redundant tell.

	E0		E1		E2		Rounds to 90% (E1)	
Learner	Seats	Cut-off	Seats	Cut-off	Seats	Cut-off	Seats	Cut-off
ε-greedy	82 ±4	**74 ±3**	71 ±5	86 ±3	72 ±4	86 ±4	never (20%)	never (50%)
UCB1	**96 ±2**	56 ±2	93 ±3	99 ±1	94 ±2	99 ±1	56k	18k
Thompson sampling	93 ±2	70 ±4	92 ±3	99 ±1	94 ±2	99 ±1	78k	19k
EXP3	93 ±2	51 ±1	81 ±4	94 ±2	84 ±4	93 ±2	141k	61k
Leak-aware UCB	**96 ±2**	56 ±2	**100 ±0**	**100 ±0**	**100 ±0**	**100 ±0**	27k	7k

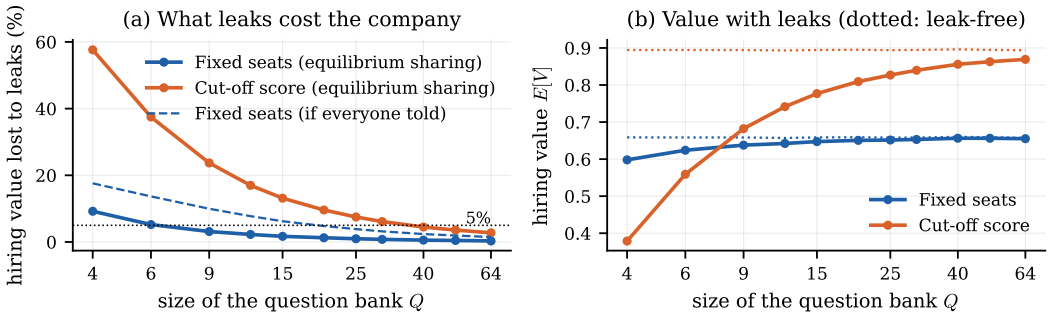


Figure 5: Company's hiring value under equilibrium sharing as the question bank grows (E2, $\alpha = 0.2$, $m = 3$ questions per interview). 95% block-bootstrap intervals (500 replicates) are narrower than the markers.

slower (81/94% and 71/86%). Results under E2 are the same within error bars.

A cut-off is easier to learn than fixed seats whenever the equilibrium is unique: the gap is larger and the same for all types. In E1 and E2, agreement under a cut-off is at least as high for every learner and significantly higher (Holm-corrected $p < 0.05$) in 8 of 10 paired tests (the exception: Leak-aware UCB, at 100% under both rules). Under fixed seats the slowest learners are the *average* types (UCB1: 99% of weak, 83% of average, 96% of strong decisions correct), although their gap is not the smallest (0.40, 0.55, 0.76 points). A plausible reason, which we did not test: a weak candidate is almost never hired, so her rewards are much less noisy.

Knowing how leaks work is worth a factor of two to three. Leak-aware UCB reaches 90% agreement after 27,000 rounds under fixed seats and 7,000 under a cut-off, against 56,000 and 18,000 for UCB1, and ends at 100% and 100%. At 20,000 rounds it is ahead of UCB1 by +13 and +12 points (E1) and +11 and +7 points (E2); all 4 differences are significant after Holm correction. Its settled profile is an exact equilibrium in 97% (seats) and 100% (cut-off) of runs, against 40% and 87% for UCB1, and it recovers the leak rate: $\hat{p} = 0.200$ in E1 and E2 (truth $m/Q = 0.2$). The help is one-sided, which is instructive: against UCB1, cumulative regret changes by −351 under a cut-off, where the best arm is "tell everyone" and each pull also informs every other arm, and by +2 under fixed seats, where the best arm for most types is silence — and silence teaches nothing about telling.

No-regret play approaches the CCE set (Figure 4c). The CCE gap falls between 10,000 and 200,000 rounds for all five learners (to 0.45–0.73 of its earlier value; Holm-corrected $p < 0.001$ in 5 of 5 tests) and ends at 0.38–0.94 percentage points. It is not yet zero: exploration is still being paid for.

A caution about "agreement". In E0 with a cut-off, agreement looks poor (51–74%). The game there has 4 equilibria that differ only in which of two friends tells the last candidate; the learners end up both telling. The residual gain from deviating is 0.10 percentage points for every learner — exactly the message cost c. They have learned everything that matters and failed only to coordinate on who saves a message.

5.4 What it costs the company (RQ4)

Leaks hurt a cut-off company about ten times more. With the pilot's bank of 15 questions, equilibrium sharing costs 1.7% of hiring value under fixed seats and 13.1% under a cut-off; across all bank sizes the cut-off loses 8–12 times more, and the bootstrap intervals are disjoint at all 11 sizes (Figure 5a). Two effects add up: under a cut-off everyone tells, and every leaked mark lifts someone over a fixed line, whereas under fixed seats only the weak tell and a leak only reorders candidates. To keep the loss below 5% the company needs $Q \geq 9$ under fixed seats and $Q \geq 40$ under a cut-off. Proposition 3 explains the shape: the leaked fraction falls like $m\,n_j/Q$.

But the cut-off is still the better rule once the bank is large enough (Figure 5b). Without leaks a cut-off hires better (value 0.89 against 0.66): it can hire three good candidates or none, where fixed seats must always hire two. Leaks eat this advantage only for small banks: fixed seats give more value up to $Q = 6$, the cut-off from $Q = 9$

Table 4: The five hypotheses fixed in the project review, and what the experiments say.

	Hypothesis	Evidence	Verdict
H1	Under fixed seats, $\alpha^\star$ is highest for the types most likely to be pivotal	$\alpha^\star = 0.10 < 0.32 < 0.38$, in the order of the chance that a tell costs one's own seat ($\ell = 0.38, 1.49, 1.60$ points)	Supported
H2	Cut-off shares more than fixed seats for every $\alpha \in [0, 0.5]$	At least as much in 188 of 189 exact comparisons, strictly more in 134; 1 exception, in E0 with several equilibria	Supported, 1 exception
H3	Fixed seats: sharing rises with β; cut-off: no effect	33%→100% (never falling) vs. constant 100%	Supported
H4	No-regret play nears the CCE set; model-based learners are faster	CCE gap falls for 5 of 5 learners; leak-aware UCB reaches 90% in 27,000/7,000 rounds vs. 56,000/18,000 for UCB1	Supported*
H5	$\Lambda_{\text{BAR}} > \Lambda_{\text{SEATS}}$, and fixed seats need a smaller Q for $\Lambda < 5\%$	8–12× larger, disjoint CIs at every Q; $Q \geq 9$ vs. $Q \geq 40$	Supported

* With the simpler leak-aware learner of Section 4; faster in agreement under both rules, lower regret only under a cut-off.

(0.78 against 0.65 at $Q = 15$). For a company the message is not "avoid cut-offs" but "a cut-off needs a question bank several times larger".

5.5 Verdict on the hypotheses, deviations and limits

All five hypotheses are supported (Table 4), H2 with one exception and H4 with a simpler learner than promised. **Deviations from the review plan:** (i) tables use 2,000 days per cell, not the pilot's 8,000; precision is instead reported per equilibrium by the bootstrap; (ii) the "model-based" learner is the leak-aware learner above and needs the friend to report how many told questions came up; (iii) mixed equilibria were sought by replicator dynamics only (no support enumeration), with the CCE programs bounding what they could do; (iv) the CCE programs were run for E1 only; (v) the complete network at $n = 6$ was handled by deviation tests, and $n = 8$ was not attempted. **Limits.** Types are coarse and rounds are independent: there is no reputation, and placed candidates do not leave the pool (a stochastic game [14]). Networks are fixed and small. There is no real leak data, so every claim is comparative (seats vs. cut-off, weak vs. strong), not a prediction of how much real candidates share. The bootstrap measures Monte Carlo error, not sensitivity to Δ, σ or K/n, which we held fixed.

6 Contribution

Member	Task assigned in the review	Where it is delivered	Status
Jash Bharat Pasad	Formulation, proofs; utility types (X2); company analysis (X4); writing, integration	Sections 2–3, Corollary 1; `utility.py`, `run_x2.py`, `run_x4.py`; Table 2, Figure 5; report	Fulfilled
Bhavay Goyal	Simulator (E0–E2, networks), payoff tables, verification checks, pipeline	`config.py`, `env.py`, `tables.py`, `checks.py`, unit tests; Table 1	Fulfilled
Chaitanya Bhoite	Level-1 solvers (BNE, mixed equilibria, CCE program, bootstrap ε); X1	`game.py`, `solvers.py`, `analysis.py`, `run_x1.py`; Figures 1–3	Fulfilled; no support enumeration
Vishnu Sinha	Level-2 learners (incl. model-based); regret and CCE metrics; X3; statistics	`learners.py`, `learning.py`, `stats.py`, `run_x3.py`; Figure 4, Table 3	Fulfilled; learner simplified

Core (verified simulator, X1–X3, H1–H4) and target (X4, H5) deliverables are complete; the stretch target $n = 8$ is not. Every number above is generated from the result files in the repository by `experiments/make_figures.py`.

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